SE3002 - Software Quality Engineering

Assignment 2: Testing Report

Student Course Registration System (black-box and white-box testing with GitHub)

Name	Roll No.	GitHub account
Junaid Javed	22P-9163	JuniJaved01
Mian Ayan	22P-9149	MianAyaan0316

Repository: https://github.com/JuniJaved01/SQE-Assignment-2

Item	Result
Test design	19 ECP + 26 BVA + 12 decision-table rows = 57 design rows; 12 exact duplicates merged
Test suites	56 black-box tests (tests/test_blackbox.py) + 10 white-box tests (tests/test_whitebox.py) = 66 tests
Initial run (original code)	16 failed, 40 passed (black-box only)
Defects found	5 distinct defects (GitHub Issues #2 to #6), all fixed and closed
Final run (after fixes, branch main)	66 tests passed; statement coverage 100%, branch coverage 100%
Cyclomatic complexity of registration_decision()	V(G) = 10 (E - N + 2 = 27 - 19 + 2 = decision nodes + 1)
Pull requests	PR #1 (black-box), PR #7 (white-box), PR #8 (defect fixes), all merged

1. Black-box test design and optimization

The requirements document (REQUIREMENTS.md, FR-01 to FR-15) was the only source of expected results. All black-box tests were designed **before** opening registration.py, and no expected value was changed to make a test pass. Every case starts from a **baseline**: a student with 0 credits, 0 dues, no completed or registered courses, and course CS201 (3 credits, capacity 30, 0 enrolled, no prerequisites). A case lists only what it changes. The full case sheets are in docs/blackbox_test_design.xlsx and docs/decision_table.xlsx.

1.1 Equivalence class partitioning (19 cases: ECP-01 to ECP-19)

Input	Requirement	Classes and test IDs
Outstanding dues	FR-04, FR-15	dues > 0 blocks (ECP-01); dues = 0 valid (ECP-02); dues < 0 invalid input (ECP-03)
Already registered	FR-05	registered in the course (ECP-04)
Prerequisites	FR-03, FR-06	none required (ECP-05, ECP-16); all completed (ECP-06, ECP-17); some missing (ECP-07, ECP-18); none completed (ECP-08)
Credit hours	FR-02, FR-07	within 18 (ECP-09); over 18 (ECP-10); invalid course credits (ECP-13); invalid current credits (ECP-14); both invalid, order of checks (ECP-19)
Availability	FR-08, FR-15	seats available (ECP-11); course full (ECP-12); invalid capacity (ECP-15)

1.2 Boundary value analysis (26 cases: BVA-01 to BVA-26)

Variable	Requirement	Boundary cases (test IDs)
Combined credits (max 18)	FR-01, FR-07	exactly 18 allowed (BVA-03, 05, 07, 23); 19 rejected (BVA-04, 06, 08, 24, 25); smallest total (BVA-01, 02, 26)
Current credits (0 to 18)	FR-02	-1 and 19 are just outside (BVA-09, BVA-10)
Course credits (1 to 4)	FR-02	0 and 5 are just outside (BVA-11, BVA-12)
Dues near zero	FR-04, FR-15	1 blocks (BVA-13); 0 allowed (BVA-14); -1 invalid (BVA-15)
Enrolled vs capacity	FR-08, FR-15	29/30 one seat left (BVA-16); 30/30 full (BVA-17); 31/30 and -1 invalid (BVA-18, BVA-19); capacity 1 empty/full (BVA-20, 21); capacity 0 invalid (BVA-22)

1.3 Decision table testing

The decision order in FR-09 is: dues, already registered, prerequisites, credit limit, availability. Because a higher-priority condition decides the outcome, the 32 possible combinations collapse to 6 rules ('-' means don't care).

Condition	R1	R2	R3	R4	R5	R6
Dues > 0	Y	N	N	N	N	N
Already registered	-	Y	N	N	N	N
Prerequisites missing	-	-	Y	N	N	N
Credits would exceed 18	-	-	-	Y	N	N
Course full	-	-	-	-	Y	N
Outcome	**DUES_ BLOCK**	**ALREADY_ REGISTERED**	**PREREQUISITES_ MISSING**	**CREDIT_ LIMIT**	**WAITLISTED**	**REGISTERED**
Test ID	DT-01	DT-02	DT-03	DT-04	DT-05	DT-06

Priority combinations (several conditions true at once, the highest priority must win): DT-07 (all five true) gives DUES_BLOCK; DT-08 (registered, prerequisites, credits, full) gives ALREADY_REGISTERED; DT-09 (prerequisites, credits, full) gives PREREQUISITES_MISSING; DT-10 (credits and full) gives CREDIT_LIMIT; DT-11 (dues and full) gives DUES_BLOCK; DT-12 (negative dues and capacity 0) raises 'Invalid dues' because dues are validated first (FR-15).

1.4 Implementation and optimization of the suite

The tests are in tests/test_blackbox.py and use assert and pytest.raises(). They are parametrized so that each executed test carries the IDs of every technique row it covers (for example **ECP-02|ECP-05|BVA-14|DT-06**), which keeps technique-to-test traceability. Ten state tests (ST01 to ST10) check that a successful registration updates the student and course, that a waitlisted student is added once, that every unsuccessful registration leaves all fields unchanged, that registration_decision() does not change state, and that withdrawal changes the right fields.

Optimization. Two tests are exact duplicates only if the function, the input and the expected output are identical; identical outputs alone do not make a duplicate. Twelve exact duplicates were found and merged, so 57 design rows run as 45 tests (plus the 11 state tests = 56 black-box tests).

Executed once as	Rows merged	Same input
ECP-02|ECP-05|BVA-14|DT-06	4 rows (3 removed)	baseline student, expect REGISTERED
ECP-12|BVA-17|DT-05	3 rows (2 removed)	enrolled 30 of 30, expect WAITLISTED
ECP-01|DT-01	2 rows	dues = 500
ECP-04|DT-02	2 rows	already registered
ECP-07|DT-03	2 rows	completed {CS101}, required {CS101, MATH101}
ECP-10|DT-04	2 rows	current 16 + course 3 credits
ECP-13|BVA-12	2 rows	check_credit_limit(10, 5)
ECP-14|BVA-10	2 rows	check_credit_limit(19, 3)

Executed once as	Rows merged	Same input
ECP-15|BVA-22	2 rows	capacity 0

Not duplicates (kept): ECP-01 (dues 500) and BVA-13 (dues 1) both give DUES_BLOCK but test a typical value and the boundary; ECP-03 (-100) and BVA-15 (-1) are likewise different inputs.

1.5 Initial execution before any fix

The first GitHub Actions run (PR #1, branch blackbox-testing) executed the suite against the original registration.py and ended with **16 failed, 40 passed**. The failures were kept as evidence (runs #1 to #12 on the blackbox-testing branch are all red) and were grouped into five defects (section 4).

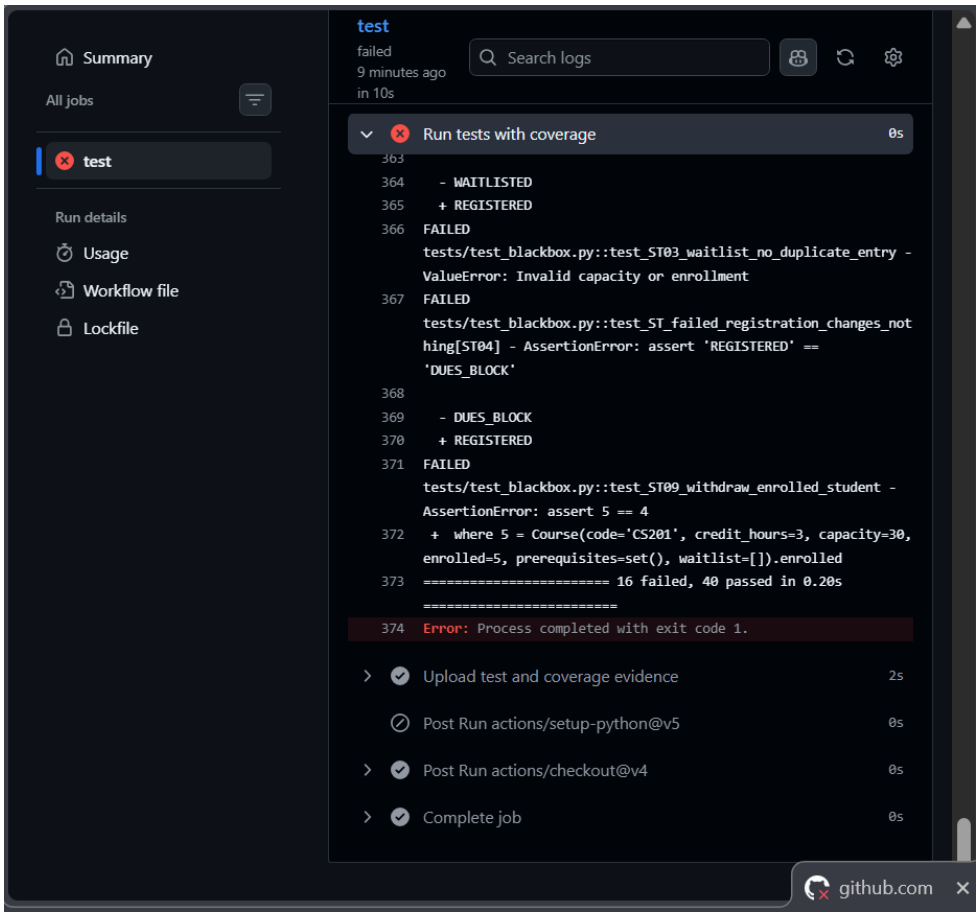


Figure 1. GitHub Actions log of the first run: 16 failed, 40 passed (exit code 1).

2. White-box analysis

2.1 Control flow graph and cyclomatic complexity

The CFG of registration_decision() is shown in Figure 2 (docs/cfg.png). **Convention for short-circuit expressions:** every atomic condition of a compound expression is its own decision node. Line 48 is *not (0 <= enrolled <= capacity) or capacity < 1*, which has three atomic conditions: A = 0 <= enrolled, B = enrolled <= capacity, C = capacity < 1. Calls to prerequisites_satisfied() and check_credit_limit() count as one decision each because their insides have their own CFGs.

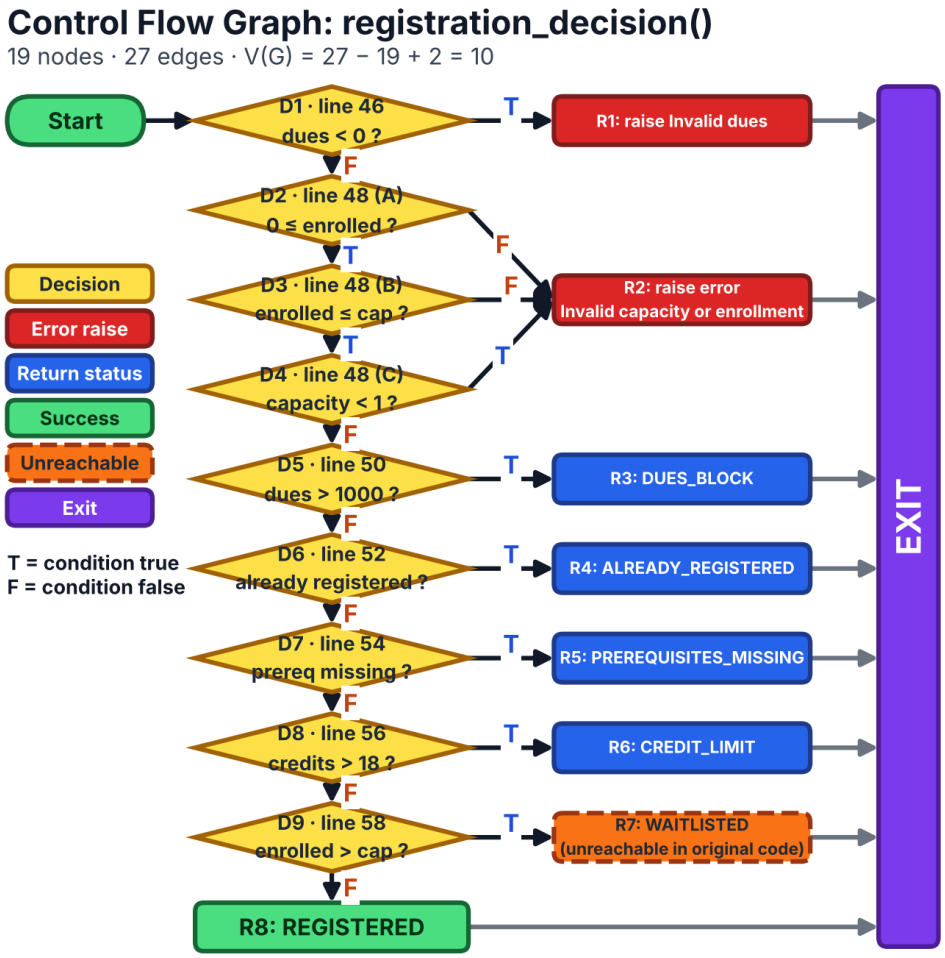


Figure 2. Control flow graph of registration_decision(): 9 decision nodes, 8 return/raise nodes, Start and Exit.

Measure	Value
Nodes N	19 (Start, 9 decisions D1-D9, 8 return/raise nodes R1-R8, Exit)
Edges E	27 (1 from Start, 18 from decisions, 8 into Exit)
V(G) = E - N + 2	27 - 19 + 2 = 10
Decision nodes + 1	9 + 1 = 10 (the two methods agree)
Line-level convention (line 48 as one decision)	7 decisions, V(G) = 8 (shown for comparison only)

2.2 Independent basis paths

Path	Route	Representative test
P1	D1 true to R1 (negative dues)	BVA-15
P2	D1 false, D2 false to R2	BVA-19 (WB04)
P3	D2 true, D3 false to R2	BVA-18 (WB05)
P4	D3 true, D4 true to R2	ECP-15 (WB06)
P5	D4 false, D5 true to R3 (DUES_BLOCK)	WB01 (added)
P6	D5 false, D6 true to R4 (ALREADY_REGISTERED)	ECP-04
P7	D6 false, D7 true to R5 (PREREQUISITES_MISSING)	ECP-08
P8	D7 false, D8 true to R6 (CREDIT_LIMIT)	ECP-10
P9	D8 false, D9 true to R7 (WAITLISTED)	WB08 (infeasible on the original code)
P10	D9 false to R8 (REGISTERED)	ECP-02 (baseline)

Finding: path P9 is infeasible in the original code. Reaching line 58 as true needs enrolled > capacity, but line 48 has already raised an error for exactly that input, so line 59 is dead code. It becomes reachable only after the fix of line 58 (defect #5). The original code therefore had 9 feasible paths; the fix restores the tenth.

2.3 Condition coverage and MC/DC

Atomic conditions of registration_decision(): c1 (line 46) dues < 0; A, B, C (line 48); c5 (line 50) dues > threshold; c6 (line 52) already registered; c7 (line 54) prerequisites not satisfied; c8 (line 56) credit limit not ok; c9 (line 58) enrolled > capacity. Each is covered both true and false by the tests, except c9 true, which is infeasible on the original code (see above).

Row	A	B	C	Decision	Test
1	F	not evaluated	not evaluated	True (raise)	WB04: enrolled -1, capacity 30
2	T	F	not evaluated	True (raise)	WB05: enrolled 31, capacity 30
3	T	T	T	True (raise)	WB06: enrolled 0, capacity 0
4	T	T	F	False (continue)	WB07: enrolled 0, capacity 30

Python short-circuits, so only four combinations are reachable. Independence pairs (masking MC/DC): **A** by rows 1 and 4 (A flips and the decision flips; B and C are masked in row 1); **B** by rows 2 and 4 (A fixed true, B flips); **C** by rows 3 and 4 (A and B fixed true, C flips). Three conditions need four tests (n + 1), and each condition has been shown to affect the outcome on its own. The second compound decision, line 70 of register_course() (*decision == WAITLISTED and student_id not in waitlist*), is covered by WB09 (both true), WB10 (second false) and WB11 (first false).

2.4 White-box tests (tests/test_whitebox.py)

Test	Target
WB01	line 50 true branch, line 51 statement, path P5 (dues 2000 gives DUES_BLOCK)
WB02	line 50 decision at the code's constant 1000; FR-04 requires any dues > 0 to block
WB04 to WB07	MC/DC rows 1 to 4 of the line-48 decision (A, B, C)
WB08	line 58 true branch, line 59, path P9 (full course gives WAITLISTED)
WB09 / WB10 / WB11	line 70 compound decision: append to waitlist / already waiting, no duplicate / registered, waitlist untouched

There are 10 white-box tests (IDs WB01, WB02, WB04 to WB11). On the original code 7 pass and 3 fail (WB02, WB08, WB09) because of the defects; after the fixes all pass.

3. Statement and branch coverage

Measured with `python -m pytest tests/ --cov=registration --cov-branch --cov-report=term-missing`. pytest-cov reports statement and branch coverage; it does not measure condition coverage or MC/DC, which were calculated manually in section 2.3.

Stage	Statements	Branches	pytest-cov total	Uncovered lines
Initial: black-box tests only, original code	54/57 = 94.7%	23/26 = 88.5%	93%	51, 59, 71
After white-box tests, original code	55/57 = 96.5%	24/26 = 92.3%	95%	59, 71
After defect fixes (PR #8), all 66 tests	58/58 = 100%	26/26 = 100%	100%	none

Gaps and how they were handled. Line 51 (the DUES_BLOCK return) was never reached because the black-box dues values (500 and 1) are below the code's wrong threshold of 1000; WB01 (dues 2000) reaches it. Lines 59 and 71 (the waitlist return and the waitlist append) cannot be reached on the original code because of defect #5; WB08 and WB09 were written for them, fail until the fix, and cover them afterwards. Reaching 100% was not the goal on its own: the coverage report exposed the unreachable path.

4. Defects, GitHub Issues and fix status

Issue	Defect (requirement)	Failing tests	Cause and fix	Status
#2	Dues below Rs. 1001 do not block registration (FR-04)	ECP-01/DT-01, BVA-13, DT-07, DT-11, ST04, WB02	line 50 used '> 1000'; changed to '> 0'	Fixed in PR #8, closed
#3	Prerequisites accepted when only some are completed (FR-03, FR-06)	ECP-07/DT-03, ECP-18	line 41 used intersection (any); changed to required.issubset(completed)	Fixed in PR #8, closed
#4	A total of exactly 18 credits is rejected (FR-01, FR-07)	BVA-03, BVA-05, BVA-07, BVA-23	line 34 used '< 18'; changed to '<= 18'	Fixed in PR #8, closed
#5	Full course is not waitlisted (FR-08, FR-12)	ECP-12/BVA-17/DT-05, BVA-21, ST02, ST03, WB08, WB09	line 58 used 'enrolled > capacity', never true after line 48; changed to '=='	Fixed in PR #8, closed
#6	Withdrawal does not decrement course enrollment (FR-13)	ST09	withdraw_course never decremented enrolled; added course.enrolled -= 1	Fixed in PR #8, closed

Sixteen failing tests map to five distinct defects, so there is one Issue per defect, not one per failing test (ST03 failed only as a knock-on of defect #5). Each Issue records the requirement ID, reproduction steps, expected and actual output, the failed test IDs and the status. The fixes were made on a separate branch (defect-fixes) after the failing evidence was preserved, and PR #8 closed all five Issues automatically.

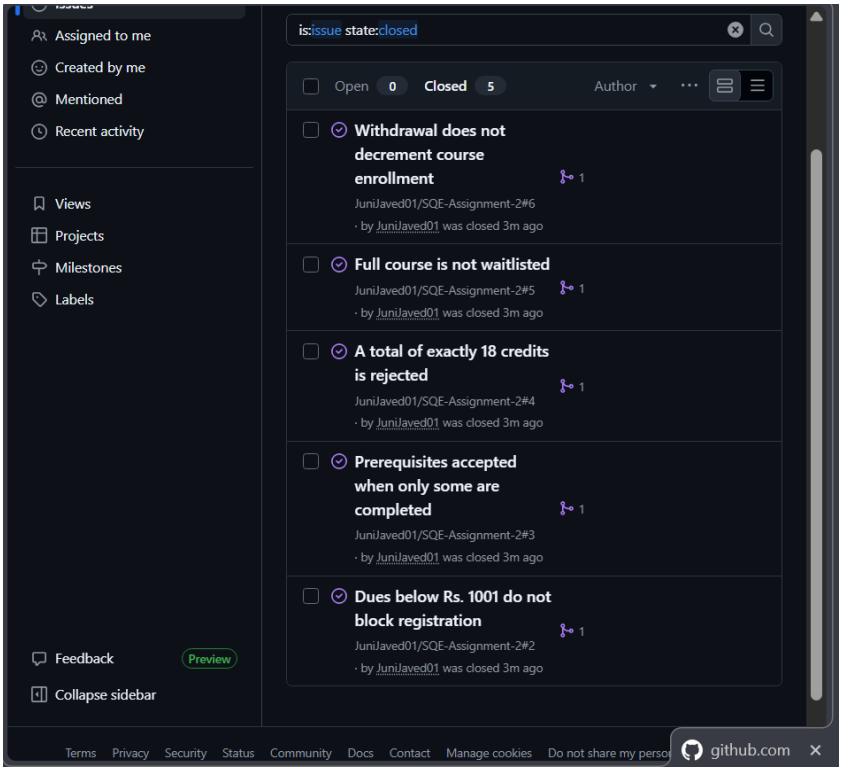


Figure 3. GitHub Issues: Open 0, Closed 5 (Issues #2 to #6), closed by PR #8.

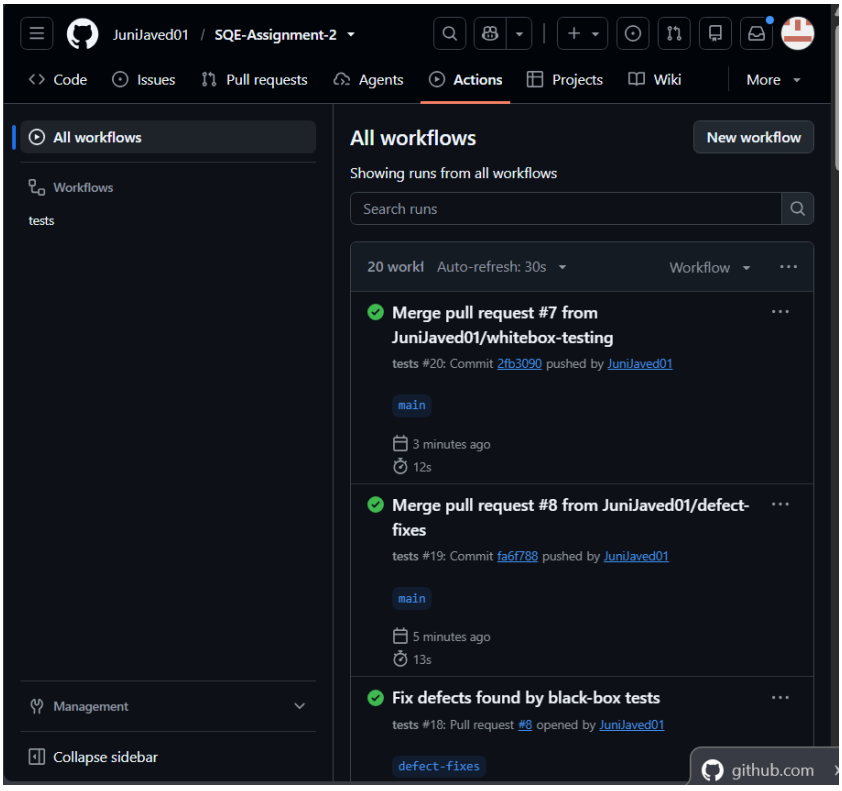


Figure 4. GitHub Actions after the fixes: runs #18 (PR #8), #19 (merge of PR #8) and #20 (merge of PR #7) are green on main.

5. GitHub repository, pull requests and contributions

Repository: https://github.com/JuniJaved01/SQE-Assignment-2. Structure: registration.py, requirements.txt, README.md, REQUIREMENTS.md, tests/test_blackbox.py, tests/test_whitebox.py, docs/ (blackbox_test_design.xlsx, decision_table.xlsx, cfg.png, testing_report.pdf), .github/workflows/tests.yml. The workflow runs pytest with coverage on every push and pull request, and uploads the test output and coverage.xml as an artifact even when tests fail.

5.1 Pull requests and reviews

PR	Branch	Author	Reviewer	Content and outcome
#1	blackbox-testing	Junaid Javed	Mian Ayan	Black-box suite, ECP/BVA/decision-table sheets, CI workflow. Mian Ayan left two review comments (why the expected value is 18, and a clearer test name); Junaid answered them. CI was red on purpose (16 failed, 40 passed). Merged.
#7	whitebox-testing	Mian Ayan	Junaid Javed	White-box tests (test_whitebox.py): MC/DC, branch and path tests. Junaid reviewed it and noted that WB08 depends on Issue #5. CI red before the fixes (3 expected failures). Merged.
#8	defect-fixes	Junaid Javed	-	Five defect fixes in registration.py and docs/cfg.png; closes Issues #2 to #6. CI green. Merged.

5.2 Member contributions

Member	Contribution
Junaid Javed (22P-9163)	Requirement analysis and black-box design (ECP, BVA, decision table, optimization), both Excel sheets, tests/test_blackbox.py, GitHub repository and Actions workflow, Issues #2 to #6, CFG, V(G), MC/DC and coverage analysis, defect fixes (PR #8), PR #1, review of PR #7, this report.
Mian Ayan (22P-9149)	White-box test suite (PR #7, tests/test_whitebox.py), review of PR #1, and collaboration on the repository.

6. Conclusion

Requirements-based tests found five real defects that the code could not reveal by itself, and the failing evidence was preserved before any fix. White-box analysis showed an unreachable path (line 59) caused by defect #5 and gave V(G) = 10. After the five one-line fixes all 66 tests pass with 100% statement and branch coverage, and the GitHub Actions run on main is green. No expected value was altered to make a test pass.